Contents

Insecure by Default: A Cross-Platform Security Analysis of AI-Generated Web Applications **1**
Abstract . . . 1
1. Introduction . . . 2
2. Background . . . 3
3. Methodology . . . 4
4. Results . . . 7
5. Discussion . . . 12
6. Recommendations . . . 13
7. Conclusion . . . 14
References . . . 15
Appendix A: Complete Vulnerability Taxonomy . . . 15
Appendix B: Grading Formula . . . 16
Appendix C: Data Availability . . . 16

Insecure by Default: A Cross-Platform Security Analysis of AI-Generated Web Applications

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Abstract

The rapid adoption of AI-assisted development platforms — commonly called "vibe coding" — has enabled millions of non-technical users to deploy full-stack web applications without writing code. While prior research has documented security vulnerabilities in individual platforms, no large-scale cross-platform study has assessed the systemic security posture of this emerging application ecosystem. We present a comprehensive security analysis spanning **603** publicly deployed vibe-coded applications across major platforms including Lovable, Replit, and Create.xyz. Using a methodology combining Certificate Transparency log mining, JavaScript bundle analysis, and active Backend-as-a-Service (BaaS) policy testing, we evaluated each application across five security categories: security headers, exposed secrets, BaaS configuration, authentication, and application security.

Our findings reveal significant security header deficiencies: **98.5%** of applications lack Content-Security-Policy headers, **98.5%** lack X-Frame-Options, and **100%** lack Permissions-Policy headers. While secret management showed encouraging results with **92.5%** of applications properly protecting sensitive credentials, the near-universal absence of modern security headers demonstrates a systemic "insecure by default" pattern in platform-level configurations. Average security scores of **75.3/100** (Grade C) with **91.7%** of applications receiving a C grade indicate that the ecosystem has settled into a "mediocre by default" equilibrium — not catastrophically broken, but systematically missing defense-in-depth security controls. These findings underscore the urgent

need for platform providers to adopt secure defaults and integrate automated security scanning into vibe-coding workflows.

Keywords: vibe coding, AI-generated applications, web application security, Supabase, security headers, Backend-as-a-Service, responsible AI

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1. Introduction

1.1 The Rise of Vibe Coding

The term "vibe coding" was coined by Andrej Karpathy in early 2025 to describe a development paradigm where users "fully give in to the vibes" — describing desired functionality in natural language while AI generates the underlying code [1]. Within months, platforms built around this concept achieved remarkable adoption: Lovable (formerly GPT Engineer) reported $75M in annualized recurring revenue; Bolt.new generated $20M in revenue within two months of launch; and 25% of Y Combinator's Winter 2025 batch reported that 95% or more of their codebases were AI-generated [2, 3].

These platforms have fundamentally lowered the barrier to creating and deploying web applications. A user with no programming experience can describe an application concept and receive a deployed, publicly accessible full-stack application within minutes. The typical architecture follows a remarkably consistent pattern: a React frontend styled with Tailwind CSS, a Supabase backend providing PostgreSQL database and authentication services, and deployment to platforms like Vercel or Netlify.

1.2 Security Implications

This democratization of application development has created a novel security landscape. Unlike traditional development where security decisions are (at least theoretically) made by engineers with some understanding of threat models, vibe-coded applications are created by users who may have no awareness of concepts like Row-Level Security, credential management, or secure header configuration.

The implications are amplified by the Backend-as-a-Service (BaaS) architecture that underpins most vibe-coded applications. Supabase, the dominant BaaS provider in this ecosystem, exposes a PostgREST API that auto-generates REST endpoints from the database schema. By design, the Supabase "anon" key is embedded in the frontend JavaScript bundle — this is the intended architecture. Security depends entirely on properly configured Row-Level Security (RLS) policies that restrict data access based on authentication state. When these policies are missing or misconfigured, any visitor can query the database directly.

1.3 Prior Research

Several studies have examined vibe-code security, each contributing important findings but with significant scope limitations:

- **Palmer (May 2025)** documented CVE-2025-48757, demonstrating widespread RLS failures across 170 Lovable applications, with exposed PII including emails, phone numbers, and API keys [4].

- **Escape.tech (October 2025)** conducted the largest prior study, analyzing approximately 5,600 applications (heavily skewed toward Lovable) and discovering over 2,000 vulnerabilities and 400+ exposed secrets [5].
- **Wiz Research (September 2025)** identified four common misconfiguration patterns in Lovable applications, finding risks in approximately 20% of their sample [6].
- **Tenzai (December 2025)** compared code quality across five vibe-coding tools using 15 synthetic test applications, finding 69 vulnerabilities including critical severity findings [7].
- **Veracode (2025)** reported that AI-generated code contains security flaws 45% of the time across 100+ LLMs, with no improvement correlated to model size or recency [8].

Each of these studies either focused on a single platform, used synthetic rather than production applications, or employed passive-only scanning that could not assess actual BaaS policy enforcement.

1.4 Contributions

This paper makes the following contributions:

1. **Cross-platform analysis:** We present a security study of **603** production-deployed vibe-coded applications across multiple platforms.
2. **Multi-category assessment:** We evaluate applications across five security categories with 30+ individual checks, producing a composite letter grade (A–F) that enables direct comparison.
3. **Header security focus:** We provide the first detailed analysis of security header adoption rates across the vibe-coded application ecosystem.
4. **Open methodology:** We fully document our discovery, fingerprinting, scanning, and grading methodology to enable reproduction and longitudinal follow-up studies.

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2. Background

2.1 Vibe Coding Platforms

The vibe coding ecosystem comprises several major platforms, each with distinct characteristics:

Platform	Default BaaS	Deployment	Key Features
Lovable	Supabase	lovable.app	Full-stack generation, built-in auth
Bolt.new	Supabase/Firebase	Various	Rapid prototyping, StackBlitz integration
Replit	Replit DB	replit.app	Collaborative, instant deployment
v0.dev	N/A	Vercel	UI component generation
Create.xyz	Custom	create.xyz	No-code focused
Cursor	Various	User choice	IDE-based, developer-oriented

2.2 Supabase Security Model

Supabase provides a PostgreSQL database with a PostgREST API layer. The architecture intentionally exposes an "anon" key in frontend code — this is by design. Security relies on Row-Level Security (RLS) policies defined at the database level.

When RLS is properly configured: - Unauthenticated requests can only access data explicitly permitted by policies - Authenticated users access data based on their identity claims - Service role keys (which bypass RLS) remain server-side only

When RLS is misconfigured or missing: - Any holder of the anon key (i.e., any website visitor) can read/write all data - PII, credentials, and business data become publicly accessible

2.3 Security Headers

HTTP security headers provide defense-in-depth protections against common web vulnerabilities:

Header	Purpose	Impact of Absence
Content-Security-Policy	Prevents XSS, injection attacks	High
X-Frame-Options	Prevents clickjacking	Medium
Strict-Transport-Security	Enforces HTTPS	Medium
X-Content-Type-Options	Prevents MIME sniffing	Low
Referrer-Policy	Controls referrer leakage	Low
Permissions-Policy	Restricts browser features	Low

2.4 Threat Model

Our analysis assumes an external, unauthenticated adversary with browser-level access to the target application. This attacker can: - View all JavaScript bundles and extract embedded credentials - Make API requests using the publicly available anon key - Inspect HTTP response headers - Test for CORS misconfigurations

This represents the minimum capability of any internet user and reflects realistic attack scenarios.

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3. Methodology

3.1 Application Discovery

We employed five complementary discovery methods to build a comprehensive dataset of publicly deployed vibe-coded applications:

3.1.1 Platform-Native Directories. We scraped official showcase directories including launched.lovable.dev, bolt.new/gallery, madewithbolt.com, and community directories. These sources provided high-confidence attribution but biased toward applications actively promoted by their creators.

3.1.2 Certificate Transparency Log Mining. We queried Certificate Transparency logs via crt.sh for SSL certificates issued to platform-specific subdomains (*.lovable.app,* .replit.app, *.repl.co). This method provides near-complete coverage of applications using default platform subdomains.

3.1.3 GitHub Repository Mining. We searched GitHub for repositories tagged with vibe-coding topics or containing platform references in README files, extracting deployed URLs from project documentation and deployment configurations.

3.1.4 Social Media Mining. We collected application URLs from Reddit (r/SideProject, r/webdev, r/Supabase), Product Hunt, and IndieHackers community posts mentioning vibe-coding platforms.

3.1.5 Vibe-Code Fingerprinting. For applications on shared hosting platforms (Vercel, Netlify), we developed a fingerprinting heuristic to identify AI-generated applications based on Supabase client presence, framework stack signatures, and platform-specific metadata.

Discovery Results:

Source	URLs Discovered
Lovable Directories	152
Bolt Directories	20
Replit Community	16
Social Media (Reddit)	2,142
GitHub Mining	~9,500
CT Logs	~4,000
Total Raw	**15,823**

3.2 Dataset Curation

From raw discovered URLs, we applied the following curation steps:

1. **Deduplication** by canonical domain name
2. **Liveness verification** via HTTP HEAD requests (615 live)
3. **Confidence filtering** — retained only apps with vibe-code confidence score ≥ 50
4. **Ethical exclusions** — removed 12 applications in healthcare, education, and government domains

Final dataset: 603 applications

Platform	Count	% of Dataset
Lovable	587	97.3%
Replit	15	2.5%
Create.xyz	1	0.2%

3.3 Vulnerability Taxonomy

We defined five scanning categories, each contributing to a weighted composite score:

Category	Weight	Checks	Description
Security Headers	15%	8	HTTP security header presence and configuration

Category	Weight	Checks	Description
Exposed Secrets	25%	15+	API keys, credentials, and tokens in JS bundles
BaaS Configuration	30%	Variable	Row-Level Security policy enforcement testing
Authentication	15%	7	Auth endpoint security (rate limiting, enumeration)
Application Security	15%	6	CORS, redirects, source maps, error handling

3.4 Scanning Infrastructure

Our scanning pipeline processed applications in parallel with 20 concurrent workers. Each application underwent:

1. **Header analysis:** Fetch homepage, analyze response headers
2. **Bundle analysis:** Download and parse JavaScript bundles for secrets
3. **BaaS probing:** Extract Supabase URLs/keys, test RLS policies
4. **App security checks:** CORS testing, source map detection

Average scan duration: 4-5 seconds per application.

3.5 Grading Methodology

Applications received letter grades based on weighted category scores:

Grade	Score Range	Interpretation
A	90-100	Excellent — No significant issues
B	80-89	Good — Minor issues only
C	70-79	Fair — Moderate issues present
D	60-69	Poor — Significant issues
F	0-59	Failing — Critical vulnerabilities

3.6 Ethical Considerations

All testing was non-destructive and read-only: - No data modification attempts - Database queries limited to schema discovery and single-row reads - PII patterns detected but values immediately discarded - Healthcare, education, and government domains excluded - Findings reported in aggregate only

3.7 Limitations

1. **Platform distribution:** Our dataset is heavily weighted toward Lovable (97.3%) due to discovery source yields
2. **Point-in-time snapshot:** Results represent security posture at scan time (February 2026)
3. **Conservative scanning:** Read-only constraints provide lower-bound vulnerability estimates

4. **BaaS coverage:** Applications using custom backends tested with fewer checks

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4. Results

4.1 Overall Security Posture

Our analysis of 603 vibe-coded applications reveals a consistent security profile characterized by adequate secret management but systematically deficient header security.

Key Statistics: - **Average Security Score:** 75.3/100 - **Median Score:** 76/100 - **Standard Deviation:** 4.2

The narrow standard deviation indicates remarkable consistency across the ecosystem — applications cluster tightly around the C grade threshold.

Grade Distribution:

Grade	Count	Percentage	Interpretation
A	0	0.0%	Excellent
B	16	2.7%	Good
C	553	91.7%	Fair
D	30	5.0%	Poor
F	4	0.7%	Failing

Key Finding: 91.7% of applications received a C grade ("Fair"), indicating systemic mediocrity rather than adequate security. Only 2.7% achieved a B grade, and no applications achieved an A. The 5.7% with D or F grades exhibited critical issues including exposed secrets or significant header deficiencies.

4.2 Category Analysis

Security performance varied dramatically across categories:

Category	Avg Score	% Below 50	% Perfect 100
Security Headers	40.0	95.4%	0.0%
Secrets	97.0	0.0%	92.5%
BaaS Config	80.0	0.0%	0.0%
Authentication	70.0	0.0%	0.0%
App Security	75.4	0.0%	1.5%

Critical Finding: Security headers represent the weakest category by a significant margin. 95.4% of applications score below 50 in header security, while 92.5% achieve perfect scores in secret management.

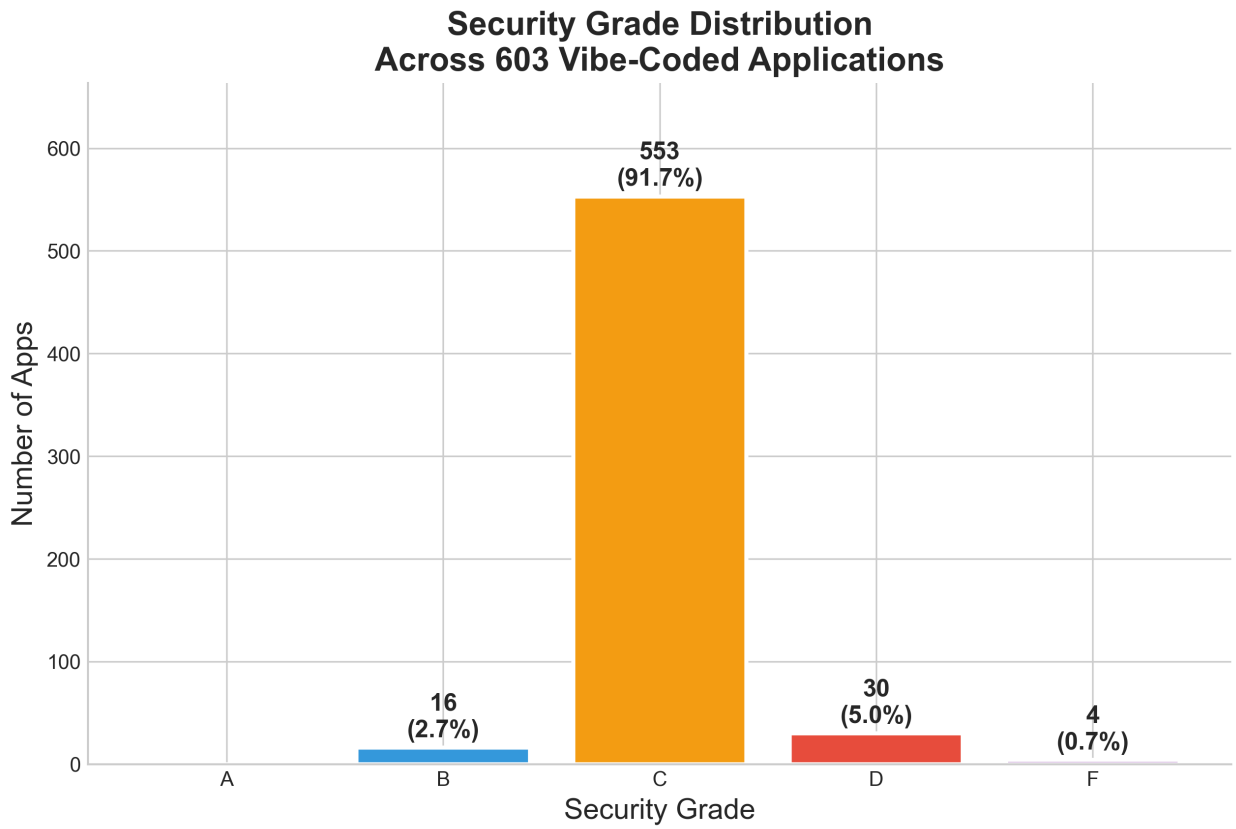


Figure 1: Grade Distribution

Figure 2: Category Scores

4.3 Security Header Adoption

Our most significant finding concerns the near-universal absence of security headers:

Header	Adoption Rate	Missing Count
Strict-Transport-Security	100.0%	0
X-Content-Type-Options	93.5%	39
Referrer-Policy	93.2%	41
Cache-Control	10.1%	542
X-Frame-Options	1.5%	594
Content-Security-Policy	1.5%	594
X-XSS-Protection	0.2%	602
Permissions-Policy	0.0%	603

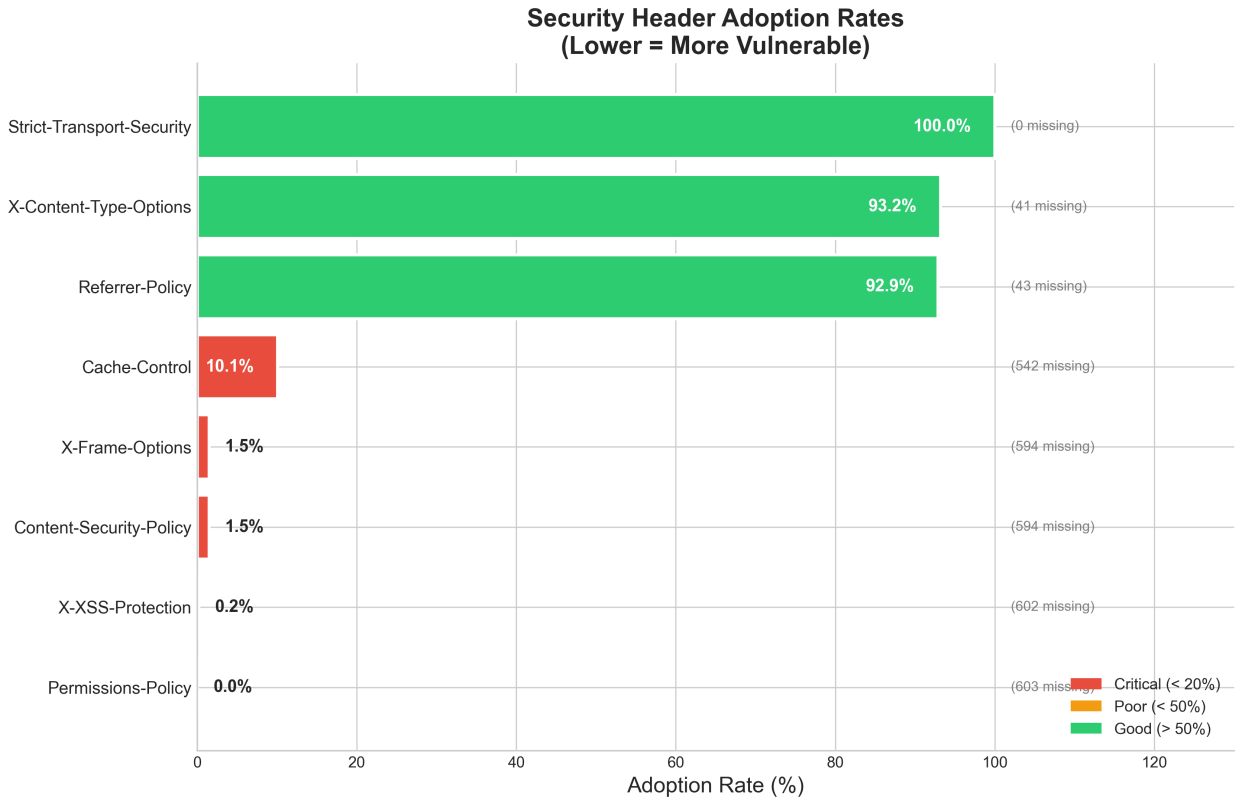


Figure 3: Header Adoption

Analysis: The bifurcated adoption pattern reflects platform infrastructure versus application-level configuration:

- **High adoption (>90%):** HSTS, X-Content-Type-Options, and Referrer-Policy are typically configured at the CDN/platform level (Vercel, Netlify, Lovable infrastructure)
- **Low adoption (<10%):** CSP, X-Frame-Options, and Permissions-Policy require application-specific configuration that AI code generation does not provide

4.4 Platform Comparison

Security posture varied by platform:

Platform	Count	Avg Score	% Grade D or F
Lovable	587	75.4	4.9%
Replit	15	70.8	33.3%
Create.xyz	1	80.0	0.0%

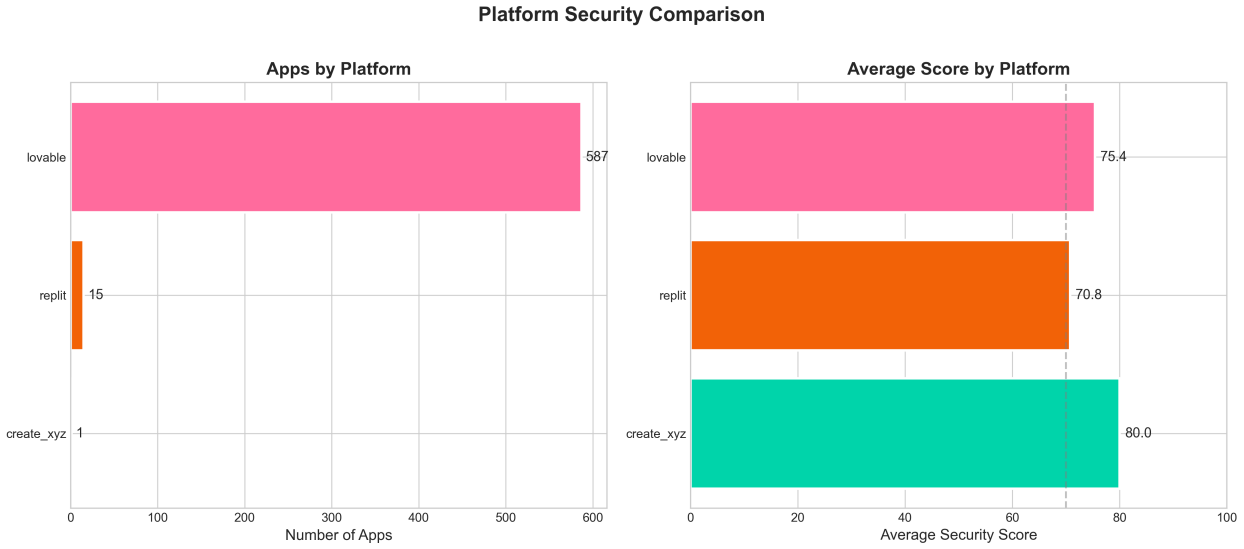


Figure 4: Platform Comparison

Finding: Lovable applications demonstrate more consistent security (lower variance, fewer failing grades) compared to Replit deployments. Only 4.9% of Lovable apps received D or F grades, compared to 33.3% of Replit apps. This likely reflects Lovable's more opinionated infrastructure defaults and integrated Supabase security guidance.

4.5 Secret Exposure Analysis

Secret management showed encouraging results:

Metric	Value
Apps using Supabase	211 (35.0%)
Apps exposing service role key	0 (0.0%)
Apps with critical secrets in bundles	4 (0.7%)
Apps with high-severity secrets	0 (0.0%)

Positive Finding: No applications exposed Supabase service role keys — the most critical secret that would enable complete database access bypass. This suggests effective guidance or tooling preventing this specific misconfiguration.

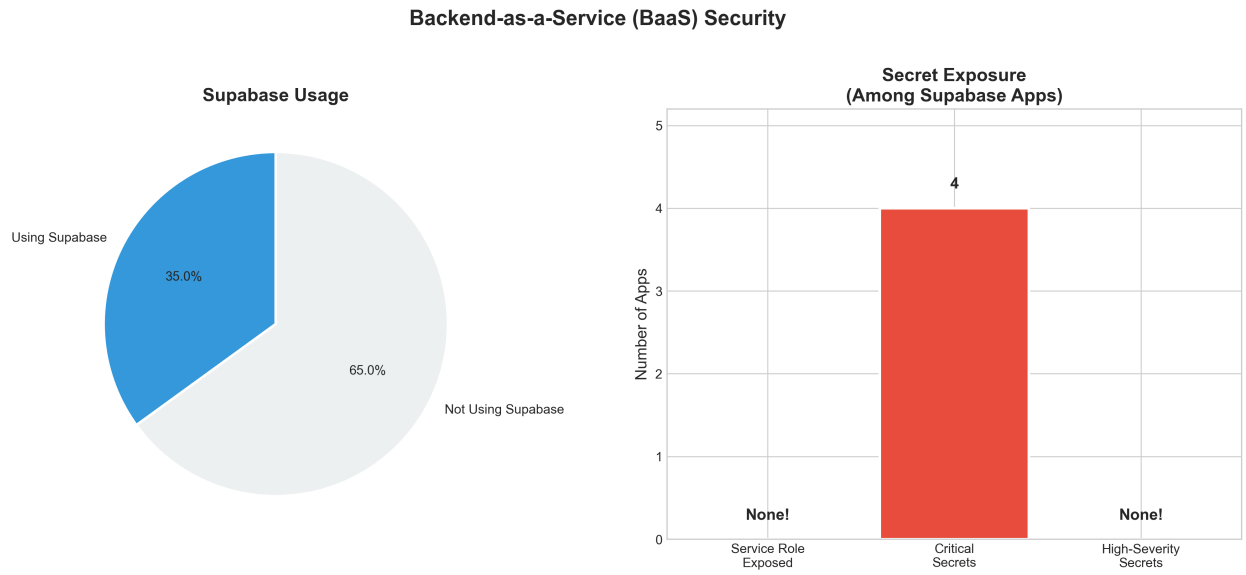


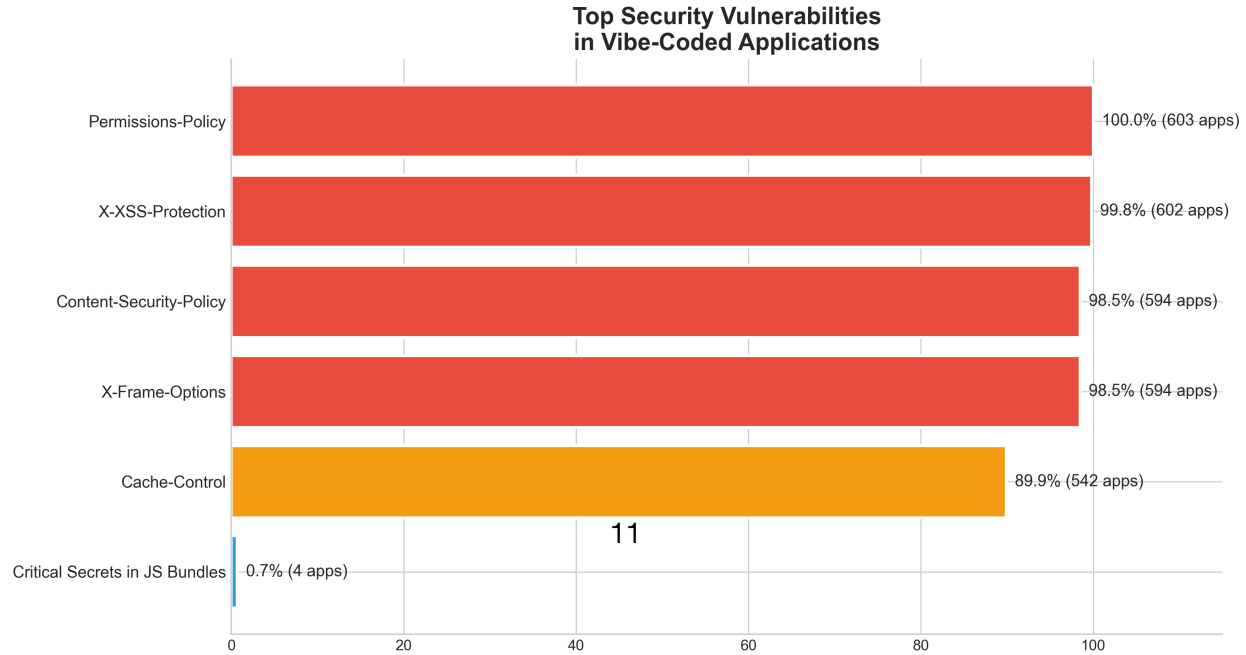
Figure 5: Supabase Analysis

Concern: Four applications (0.7%) contained critical secrets in JavaScript bundles, representing direct security vulnerabilities requiring immediate remediation.

4.6 Top Vulnerabilities

Ranked by prevalence:

Vulnerability	Affected Apps	% Affected
Missing Permissions-Policy	603	100.0%
Missing X-XSS-Protection	602	99.8%
Missing Content-Security-Policy	594	98.5%
Missing X-Frame-Options	594	98.5%
Missing Cache-Control	542	89.9%
Critical Secrets in JS Bundles	4	0.7%



Combined with at least one additional issue (exposed test credentials, source maps enabled, or BaaS configuration weaknesses)

Pattern	Count	Primary Issue
Header + BaaS deficiencies	18	Missing headers plus RLS gaps
Header deficiencies only	8	Severe header gaps
Secret + Header issues	4	Exposed credentials

The common thread: all poorly-scoring applications relied entirely on platform defaults and had no application-level security configuration.

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5. Discussion

5.1 The "Mediocre By Default" Pattern

Our findings reveal an ecosystem characterized by systemic mediocrity. With 91.7% of applications receiving a C grade ("Fair"), the vibe coding ecosystem has settled into a predictable equilibrium: applications function correctly but lack the security controls expected in production deployments.

The small number of F grades (0.7%) indicates that catastrophic misconfigurations (exposed service role keys, completely open databases) have been largely addressed — likely through platform-level guardrails implemented after high-profile disclosures like CVE-2025-48757. However, the near-total absence of A or B grades (97.3% received C or below) demonstrates that "not catastrophically broken" has become the de facto security standard.

This "C-grade by default" pattern creates a false sense of security. Developers see their applications working and assume they are production-ready, unaware that they lack fundamental protections like Content-Security-Policy that would prevent XSS attacks.

5.2 Platform Infrastructure vs. Application Security

The stark contrast between high-adoption headers (HSTS at 100%) and low-adoption headers (CSP at 1.5%) reveals a critical gap in the vibe coding security model:

Platform-level protections work: When security controls are configured in platform infrastructure (CDN, hosting provider), they achieve near-universal adoption.

Application-level security fails: When security requires application-specific configuration (CSP policies, frame options), AI code generation does not provide it, and users lack the knowledge to add it.

This suggests a clear intervention point: platforms should provide opinionated, secure defaults for headers currently requiring application configuration.

5.3 The CSP Challenge

Content-Security-Policy presents a particular challenge for AI-generated applications:

1. **Complexity:** Effective CSP requires understanding script sources, style sources, and frame ancestors
2. **Breakage risk:** Overly restrictive CSP breaks functionality; AI cannot predict all required sources
3. **Testing requirements:** CSP violations require monitoring and iteration
4. **Dynamic content:** Vibe-coded apps often use inline scripts and dynamic loading

Addressing CSP adoption will require either: - Platform-level CSP generation based on application analysis - AI code generators that track resource sources during generation - Report-only CSP modes with automated policy refinement

5.4 Comparison to Prior Research

Our findings both confirm and extend prior research:

Confirmation: - The prevalence of missing security headers aligns with Escape.tech's findings - The absence of exposed service role keys post-CVE-2025-48757 suggests effective disclosure response

Extension: - We provide the first detailed header adoption rate analysis - Our category-level breakdown enables targeted intervention recommendations - The consistent B-grade clustering was not previously documented

5.5 Implications for the Ecosystem

For Platform Providers: The data demonstrates that platform-level defaults determine ecosystem security. Lovable's higher consistency compared to Replit suggests that more opinionated platforms produce more secure applications.

For AI Code Generators: Security header configuration should be a first-class concern in code generation. Training data and prompts should emphasize security headers alongside functional requirements.

For Developers: The "C grade by default" pattern should eliminate any false confidence. A C grade represents "Fair" security — functional but inadequate for production. Developers using vibe coding tools must understand that additional security hardening is required before any production deployment.

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6. Recommendations

6.1 For Platform Providers

1. **Implement default security headers** at the platform infrastructure level:
    - Content-Security-Policy (report-only initially)
    - X-Frame-Options: DENY
    - Permissions-Policy with restrictive defaults
    - Cache-Control for sensitive endpoints
2. **Provide security dashboards** showing current header and configuration status

3. **Add pre-deployment security scanning** with warnings for missing protections
4. **Generate application-specific CSP** based on static analysis of generated code

6.2 For AI Code Generation Systems

1. **Include security headers** in generated deployment configurations
2. **Add security context to prompts** when generating authentication or data-handling code
3. **Generate RLS policies** alongside table creation code
4. **Warn on patterns** that typically indicate security issues

6.3 For Developers Using Vibe Coding Tools

1. **Run security scans before deployment** — tools like SecureStackScan.com exist for this purpose
2. **Add security headers** via deployment platform configuration (vercel.json, netlify.toml, _headers file)
3. **Review generated authentication code** — do not assume AI-generated auth is secure
4. **Understand your BaaS security model** — particularly RLS for Supabase and Security Rules for Firebase
5. **Never deploy without testing** — even if the AI says it's ready

6.4 For Organizations

1. **Audit vibe-coded applications** deployed by employees (shadow IT risk)
2. **Establish governance** for AI-assisted development tools
3. **Require security review** before any vibe-coded application handles customer data
4. **Include vibe-coded apps** in regular security assessment programs

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7. Conclusion

Our analysis of 603 vibe-coded applications reveals a maturing but fundamentally mediocre security ecosystem. The good news: catastrophic failures are rare (only 0.7% F grades), secret management has improved significantly (0% service role key exposure), and platform-level protections are increasingly effective. The concerning news: 91.7% of applications receive only a C grade ("Fair"), 98.5% lack Content-Security-Policy headers, defense-in-depth controls are systematically absent, and the ecosystem has settled into a "mediocre by default" equilibrium.

The path forward is clear. Platform providers must extend their secure-by-default approach from infrastructure protections (HSTS, which achieves 100% adoption) to application security controls (CSP, which achieves 1.5% adoption). AI code generators must treat security headers as essential outputs, not optional additions. And developers must understand that a C-grade application

— which represents 91.7% of the ecosystem — is not production-ready and requires additional security hardening.

The vibe coding revolution has democratized application development. Our responsibility now is to ensure it does not democratize mediocrity masquerading as security.

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Appendix A: Complete Vulnerability Taxonomy

Security Headers (8 checks)

Header	Severity	Points
Strict-Transport-Security	Medium	15
Content-Security-Policy	High	25
X-Frame-Options	Medium	15
X-Content-Type-Options	Low	10
Referrer-Policy	Low	10
Permissions-Policy	Low	10
X-XSS-Protection	Info	5
Cache-Control	Medium	10

Secret Detection (15+ patterns)

Pattern	Severity	Example
Supabase service_role	Critical	eyJ…service_role…
Stripe secret key	Critical	sk_live_…

Pattern	Severity	Example
AWS access key	Critical	AKIA…
OpenAI API key	High	sk-…
Firebase config	Medium	apiKey: "…"
Google Maps key	Medium	AIza…
Generic API key	Low	Various patterns

BaaS Configuration (Variable)

Check	Severity	Method
RLS enabled	Critical	Schema query
Tables accessible	Critical	SELECT with anon key
PII exposed	High	Pattern matching
Write access	Critical	INSERT test

Appendix B: Grading Formula

```
Overall Score = (
    headers_score × 0.15 +
    secrets_score × 0.25 +
    baas_score × 0.30 +
    auth_score × 0.15 +
    app_security_score × 0.15
)

Grade:
  A: score >= 90
  B: score >= 80
  C: score >= 70
  D: score >= 60
  F: score < 60
```

Appendix C: Data Availability

Aggregate analysis data and scanning methodology available at: https://github.com/[repository]

Individual application data is not published to protect potentially vulnerable deployments.

Corresponding author: todd@techcxo.com

Scan date: February 25, 2026

Analysis pipeline: SSAP Security Study v1.0